Graham A. Neely

LinkedIn | (303) 514-2055 | graham_neely@brown.edu | Portfolio

EDUCATION

Brown University **Providence, RI**
BSc Mechanical Engineering, BA Applied Math | **GPA:** 4.00/4.00 *Expected May 2028*
Colorado Academy *High School Diploma (May 2024)* | **GPA:** 4.39/4.00 (Top 3) | **SAT:** 1570 **Denver, CO**

AEROSPACE/ENGINEERING EXPERIENCE

T-REX: Time Resolving Explorer Satellite | Black Hole Observation Research Lab **Providence, RI**
Co-founder and Undergraduate Engineering Lead *Sept. 2024 – Present*

- Co-founded a research lab designing and building a small form factor, inexpensive next-generation LEO black hole observation satellite. Satellite will enable cutting-edge time resolution of celestial targets, along with extending existing VLBI baselines.
- Won selective Brown SPRINT Summer Research Grant (2026) to fund 3D modeling satellite bus, prototyping, and orbital simulations.
- Advanced to the final round of NASA Innovative Advanced Concepts (NIAC) Phase I for full concept research funding..
- **Leading construction of 1m VLBI Radio Telescope in collaboration with MIT Haystack Observatory**
  - Architected parabolic receiver system to capture targets at 1420 MHz such as the Crab Nebula, and digitization system to correlate data. Fabricated automated tracking mount with radio interference safe design and mounted telescope components.
- **Researched, Modeled, and Simulated Spacecraft Bus and Radio Antenna design**
  - Designed laser data downlink scheme capable of 100 Gbps space-to-earth for high-volume VLBI observation. Presented to MIT Lincoln Lab and entered preliminary talks for the supply of TBIRD data downlink satellite derivative.
  - Optimized radio antenna diameter and surface RMS error versus cost for minimum required integration time for black hole, AGN, and pulsar targets. Drafted spacecraft bus design with researchers and engineers from Princeton and JPL.
  - Simulated radio receiver thermal performance and requirements for 86-150 GHz deep space radio observation.

Brown Space Engineering (BSE) | Student-Run Satellite Development Organization **Providence, RI**
Research and Development Lead, Robotics Test Lead *Sept. 2024 – Present*

- Elected head of R&D team; designed and manufactured a high altitude 115,000 ft, 2 kg payload weather balloon as a near space component testing platform for solar panels, sensors, and deployable arm for BSE's CubeSat PVDX.
- Constructed PVDX CubeSat to test novel Perovskite solar panels and passed NASA Vibration check for launch on Falcon 9 in May..
- Design and construct challenge and testing environments for undercarriage assembly validation, wheel performance, and future mission use cases for the scale NASA Curiosity Rover constructed and coded by the BSE robotics team.
- Onboard new team members and teach the basics of engineering, including CAD, Python, 3D printing, and laser cutting.

Brown Rocketry | Student-Run Rocketry Team **Providence, RI**
Avionics Lead *Sept. 2024 – Present*

- Constructed avionics payload including gyro, altimeter, landing siren, and camera using Raspberry Pi for high-powered rocketry. Soldered components and programmed data collection on RPi. Rocket was successfully launched and recovered.
- Researched optimal fin, nose, and body shape for optimal flight performance and implemented the designs with molded fiberglass.

Turbo Jet Engine Design and Construction | Independent Engineering Study **Denver, CO**
Designer and Fabricator *Summer 2024*

- Modeled thermal performance of various combustion chambers, coolant loops, and housing designs. Machined and welded dual-barrel combustion chamber and 3D printed flanges in steel. Designed and installed an oil cooling loop through the turbo.
- Ran stress tests of the engine, finding points of overheating and turbo freeze. Achieved full self-sustaining combustion with propane and air. Completed the process from initial research and CAD design to manufacturing, assembly, and testing in six months. (Article)

OTHER EXPERIENCE

The Wheeler School | K-12 Instruction **Providence, RI**
Robotics and Space Science Instructor *2025 – Present*

- Designed and taught a Space Exploration course covering space exploration history from the trials of Galileo to the ISS.
- Instructed robotics construction and coding from the ground up to a small group and mentored a FIRST LEGO League team.

Schoolhouse.world | Online Academic Instruction **Denver, CO**
Volunteer SAT Tutor *Jan. 2023 – May 2023*

- Volunteer tutor on pilot project to increase access to SAT prep materials, invited by the CollegeBoard based on test scores.

AWARDS/ACCOMPLISHMENTS

Publications: Navigation Challenges for T-REX (Abstract), Time-Resolving Sgr A* at 86 GHz via Low-Earth Orbit VLBI, T-REX: Time-Resolving Explorer Satellite
Invited Presenter: AAS 247th Conference, GWPAW Conference**,** ION Pacific PNT, IAU Symposium No. 405, Falling Walls Lab

SKILLS & INTERESTS

Technical Skills: Fusion 360, SolidWorks, Onshape, Python, MATLAB, Microsoft Office, Adobe Creative Suite, Agentic AI Workflows
Language: Bilingual proficiency in English and Spanish, Intermediate proficiency in Mandarin Chinese
Interests: Space, Tinkering, Investing, Reading, Debate, Public speaking (Commencement), Ceramics, Skiing, Sailing, Ultimate Frisbee